Sai Krishna Vaddeboina

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PROFESSIONAL SUMMARY

MS in Financial Technology candidate (graduating May 2026, GPA 3.81) building production-grade fintech systems across machine learning, blockchain, and business intelligence. Led a sponsored industry capstone with Masabi × Gemsen, deploying a 4-layer Hidden Markov Model fraud-detection pipeline that surfaced 15,195 novel fare evaders missed by existing rule engines (20.5% novelty rate, 2.9M+ events). Delivered an AutoML loan-decision analyzer on Google Vertex AI (ROC AUC 0.936) and a full-stack Solidity + Vue.js property dApp on Ethereum Sepolia. Strong in Python, SQL, Tableau, Solidity, and U.S./EU fintech regulation (GENIUS Act, MiCA, Basel III).

EDUCATION

Worcester Polytechnic Institute (WPI), Worcester, MA **Aug 2024 – May 2026**

Master's in Financial Technology (FinTech), GPA: 3.81/4.0

Relevant Coursework: AI & Its Business Applications, Financial Analytics, Cryptocurrencies & Financial Markets, Business Application of Blockchain Tech, Business Applications in Machine Learning, Business Intelligence, Data Management for Analytics, Data Visualization, Marketing Research, MS Capstone Project

Noble Degree & PG College, Hyderabad, India **Dec 2020 – May 2024**

Bachelor of Commerce in CA, CGPA: 8.0/10.0

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, scikit-learn, hmmlearn, Matplotlib), SQL, Solidity, JavaScript, R, Jupyter Notebook

Machine Learning & AI: Hidden Markov Models, K-Means, Random Forest, XGBoost, Logistic Regression, feature engineering, cross-validation, Google Vertex AI (AutoML), SHAP, UMAP

Data Analysis & Visualization: Tableau, Power BI, Excel, D3.js, NVivo 12 Plus, dashboard design, conjoint analysis

Blockchain Development: Truffle, OpenZeppelin, Web3.js, Chainlink Oracles, Ethereum (Sepolia testnet), Hyperledger Fabric, smart-contract testing (Mocha/Chai)

Web Development: Vue.js, Express.js, MongoDB, REST APIs, HTML/CSS/JavaScript, Node.js

Cloud & DevOps: AWS S3, Microsoft Azure, Google Cloud, Docker, Git/GitHub, CI/CD basics, version control

FinTech Domain: Credit risk, fraud detection, regulatory compliance (GENIUS Act, MiCA, Basel III), CCPA/COPPA, FERPA/HIPAA, stablecoin architecture, earnings analysis

Soft Skills: Cross-functional collaboration, technical communication, stakeholder presentation, project management, qualitative interviewing

Office & Productivity: Microsoft Office Suite (Word, Excel, PowerPoint), Google Workspace, Bloomberg Terminal

Operating Systems: Windows, macOS, Linux

EXPERIENCE

Industry-Sponsored Capstone Researcher · Masabi × Gemsen × WPI **Jan 2026 – May 2026**

Worcester, MA · Industry-Sponsored Project

- Engineered a four-layer machine-learning pipeline (rules + 9-state Hidden Markov Model + composite anomaly score + K-means clustering) to detect inspector-triggered fare-evasion on the Massachusetts Bay Transportation Authority (MBTA) commuter-rail network.
- Trained the HMM on 2,904,144 activation–scan events across 221,382 unique riders spanning April 2017 – January 2026; selected 9 latent states by Bayesian Information Criterion across 70 Baum-Welch fits (~11.4 hours of training).

- Identified 15,195 evaders missed entirely by Masabi's existing rule engine — 20.5% novelty rate, exceeding the project threshold negotiated at inception with the industry sponsor.
- Performed feature engineering on 86 features across 14 logical groups, including a vehicle/minute-level inspector-presence proxy whose reactivity-gap signal reaches its theoretical maximum (1.0) for the most evasive subpopulation.
- Segmented the suspicious population into six operationally distinct enforcement clusters (silhouette 0.183, K-means K=6) and authored a 60-rider human-validation review sheet with event-level evidence delivered to the industry partner.
- Presented findings via final report (38 pages), poster, and stakeholder presentation; collaborated with a five-person cross-functional team and the partner organization.

PROJECTS

Data Visualization: U.S. Cargo-Theft Surprise Index **Jan 2026 – May 2026**

Tools: *D3.js, Python (pandas, matplotlib), HTML/CSS/JS, Git/GitHub, FBI NIBRS data, Census 2024*

- Designed and shipped an interactive web dashboard exploring 2024 cargo-theft surprise (rate vs. population) across all U.S. states with funnel, ranked-bar, scatter, and choropleth views.
- Built a reproducible Python pipeline that ingests FBI NIBRS and Census XLSX data and produces both data files and report figures.
- Authored the analytical report, prospectus, and screencast script; published the project as a multi-page d3.js website.

User Experience Applications: Argus Eye-Tracking Workflow & Mobile-Game Ad Study **Jan 2026 – May 2026**

Tools: *Argus ETVision / ETAnalysis, Excel, statistical analysis (paired & two-sample t-tests)*

- Authored the data-collection manual and click-by-click analysis playbook for the Argus eye-tracking system covering calibration, AOI definition, fixation/dwell computation, pupil-diameter cognitive-load measurement, and blink detection.
- Designed a between-subjects ad-placement study (n=20, top vs. bottom mobile-game ads) with eye-tracking attention metrics and Likert enjoyment as dual dependent variables.
- Co-led a within-subjects usability study (n=31) comparing search vs. chatbot retrieval; chatbot rated significantly easier ($t(29) = -2.30$, $p = .03$).

AI & Its Business Applications: LoanDesk AI – Loan Decision Analyzer **Aug 2025 – Dec 2025**

Tools: *Google Vertex AI (AutoML Classification), Gemini 2.5 Flash, Python, JSON/REST*

- Built and deployed an AI-powered loan pre-qualification tool on Vertex AI achieving ROC AUC 0.936, PR AUC 0.931, F1 0.885 over 15+ applicant features.
- Designed an interactive web dashboard supporting individual applications and bulk CSV batch processing with explainable approval/rejection confidence scores.
- Integrated SHAP-style explainability and bias auditing across income, geography, and age groups — ensuring fairness and human-in-the-loop review for credit decisions.

Cryptocurrencies & Financial Markets: Stablecoin Regulation & Architecture Analysis **Aug 2025 – Dec 2025**

Tools: *Bloomberg Terminal, Hyperledger Fabric (Docker), regulatory frameworks (GENIUS Act, MiCA, Basel III)*

- Authored a 5-essay technical analysis covering the CFTC tokenized-collateral decision, GENIUS Act, narrow-banking risks, blockchain architecture choices for regulated stablecoins, and AI-RegTech.
- Argued (with comparative tables) that a private permissioned blockchain best satisfies GENIUS-Act compliance, scalability, and consumer-protection requirements while preserving cross-border interoperability.
- Ran in-class Hyperledger Fabric exercises on Docker — peers, orderers, certificate authorities — connected to Bloomberg market-data flows.

Teaming & Organizing for Innovation: Organizational Behavior Case Studies **Aug 2025 – Dec 2025**

Tools: *Edmondson's psychological-safety framework, IDEO design thinking, DiSC assessment, qualitative interviewing*

- Co-authored solutions to six organizational-behavior scenarios drawn from real internship experiences (innovation bottlenecks, cross-cultural collaboration breakdowns, retention loops in sales onboarding).
- Applied Edmondson, Druskat & Wolff, Brown, and Stone/Patton/Heen frameworks to design phased interventions for psychological safety and learning organizations.
- Conducted a structured leadership interview with a Financial Manager at Fidelity Investments to study creativity and innovation under regulatory constraints.

Blockchain-based Property Management System (Decentralized Application) **Jan 2025 – Apr 2025**

Tools: *Solidity, MongoDB, Express.js, Vue.js, Truffle, OpenZeppelin, Chainlink, Docker, Web3.js, Sepolia testnet*

- Developed a full-stack decentralized application for real-estate transactions deployed on the Sepolia testnet (contract 0xd761…1191).
- Implemented role-based access control (Development Authority / Approvers / Buyers-Sellers), multi-signature transfer approvals, and a Chainlink oracle for ETH/USD price validation.
- Dockerized frontend, backend, and MongoDB; passed all 8 smart-contract function tests (Truffle + Mocha + Chai).

Predicting Student Depression Risk (Machine Learning Project) **Jan 2025 – Apr 2025**

Tools: *Python (pandas, scikit-learn, matplotlib), Random Forest, Logistic Regression, XGBoost, feature engineering*

- Built and validated a Random-Forest predictive model on 27,000+ student records to identify early depression risk; selected as best performer over Logistic Regression and XGBoost.
- Achieved 87% accuracy, 88% recall, F1 = 0.87, ROC AUC = 0.92 with stratified 70/30 split and 5-fold cross-validation.
- Delivered ethical, FERPA/HIPAA-compliant business recommendations and a tiered intervention framework for higher-education counselors.

Earnings Announcements & Stock Price Movements Analysis **Jan 2025 – Apr 2025**

Tools: *Python (pandas, matplotlib), Jupyter Notebook, Yahoo Finance API, Excel, OLS regression*

- Analyzed how earnings surprises correlate with 1-day, 5-day, and 20-day stock-price movements across five sectors (NVDA, GE, JPM, SPG, PFE) over a 10-year window.
- Performed market-adjusted OLS regression, volatility, trading-volume, and Bollinger-band analyses; identified PEAD effects in tech and asymmetric loss-aversion patterns in industrials.

Business Intelligence: Myntra USA Market Expansion **Aug 2024 – Dec 2024**

Tools: *Tableau, Power BI, Microsoft Azure, SQL, conjoint analysis*

- Conducted BI analysis on Myntra's US-market entry strategy — segmented Gen-Z and Millennial customers, modeled CCPA/COPPA compliance, and produced a Kotter-style change-management plan.
- Built two interactive Tableau dashboards (Sales Management + Product Performance) tracking revenue contribution, brand performance, payment methods, and delivery-cancellation correlations.
- Ran a conjoint analysis identifying premium pricing, casual styles, and local brands as the highest-utility product attributes for the target segment.

Data Management & Analytics: Amazon Sales Data Analysis **Aug 2024 – Dec 2024**

Tools: *MySQL, SQLite, Tableau, Excel, Python (Pandas, scikit-learn, Matplotlib)*

- Analyzed 10,000+ Amazon sales records to uncover pricing trends, customer behavior, and category-level engagement.
- Applied K-Means clustering, Random Forest classification (87% accuracy), linear regression ($R^2 = 0.72$), Apriori association rules, and PCA dimensionality reduction.
- Built interactive Tableau dashboards covering category performance, discount strategy, and review trends.

Marketing Research: Student Class Participation Personas **Aug 2024 – Dec 2024**

Tools: *Qualtrics, Otter.AI, NVivo 12 Plus, Excel, qualitative coding*

- Conducted 7 student + 3 professor in-depth interviews; transcribed via Otter.AI and applied open + axial coding to develop participation personas.
- Synthesized findings into recommendations for improving in-person, synchronous-online, and asynchronous-online classroom engagement.
- Delivered a final infographic and short documentary video alongside a multi-phase research plan.

CERTIFICATIONS

- Microsoft Office Intermediate Program – Word, Excel, PowerPoint
- SAP Finance & Controlling (FI-CO) Power User Course Certificate
- Data Analytics and Visualization Job Simulation Certificate – Accenture
- International Conference of CA Students – ICAI, Hyderabad (Dec 2-3, 2022)